TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Process Digital Twin for Vacuum-Integrity–Qualified Additively Manufactured Composite Tooling for Autoclave Applications**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Smart Manufacturing*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Additive manufacturing equipment*	*Digital manufacturing*	*Process simulation / digital twin*
Concept of Proposed Technology	**Core Concept** The proposed technology aims to enable vacuum-integrity–qualified additively manufactured composite tooling for autoclave-based composite manufacturing through the development of a process-level digital twin driven by machine cards and material cards. In current industrial practice, additively manufactured composite tooling frequently fails autoclave qualification due to loss of vacuum integrity caused by insufficient bead fusion, inconsistent bead overlap, and interconnected porosity generated during the printing process. These defects originate from the printing stage and cannot be reliably corrected through post-processing alone, making upstream process control essential.			

	Key Functions and Potential Applications To address this challenge, the proposed digital twin framework captures the coupled thermo-flow and interlayer fusion behavior occurring during material extrusion and deposition. Material cards represent extrusion flow behavior, thermal properties, and fusion characteristics of composite feedstock materials, while machine cards describe printer-specific thermal boundaries, deposition behavior, and process variability. By integrating flow simulation, thermal simulation, and interlayer bonding models, the digital twin predicts how printing parameters, such as nozzle temperature, printing speed, bead overlap, layer time, and stacking temperature, govern bead geometry, fusion quality, and the formation of leakage pathways [1–4]. This simulation-guided framework enables predictive optimization of printing parameters prior to fabrication, significantly reducing reliance on trial-and-error approaches. As a result, additively manufactured composite tooling with stable vacuum integrity can be produced consistently, enabling reliable operation under autoclave pressure and temperature conditions and expanding the industrial applicability of additive manufacturing for high-performance composite tooling. The proposed process digital twin framework builds on established studies of extrusion deposition additive manufacturing that link printing conditions, bead geometry, fiber orientation, and deformation behavior [1–4].
Rationale for Joint R&D Project	**Critical Technical Barriers** Although large-scale additive manufacturing offers clear advantages for composite tooling, including reduced cost, shorter lead time, and greater design flexibility, its adoption in autoclave-based composite manufacturing remains limited due to unreliable vacuum integrity of printed tools. Achieving stable vacuum performance requires an integrated understanding of material-level melt flow and interlayer fusion behavior, machine-level thermal and deposition characteristics, and their coupled interaction during printing under industrial conditions. In current practice, optimal printing parameters for vacuum-integrity-qualified tooling are not provided by either machine manufacturers or material suppliers. Machine vendors characterize hardware performance but lack insight into material-specific fusion behavior, while material suppliers characterize materials without accounting for printer-specific thermal histories and deposition strategies. As a result, end users must rely on extensive trial-and-error experimentation, leading to long development cycles, high cost, and inconsistent outcomes. Prior research has demonstrated that printing outcomes are governed by tightly coupled interactions between material behavior, deposition physics, and machine-specific thermal histories that cannot be optimized through vendor-provided parameters alone [1–3]. **Why Global Collaboration?** The coupled thermo-flow and interlayer bonding phenomena cannot be corrected through post-processing and require predictive control at the printing stage. Developing such capability demands advanced multi-physics modeling, systematic calibration, and validation under autoclave-scale conditions. The Composites Manufacturing and Simulation Center (CMSC) at Purdue University possesses internationally recognized expertise in simulation-driven composite manufacturing, digital twin development, and additive manufacturing of composite tooling, enabling rapid risk

	reduction and accelerated technology maturation. By combining Purdue CMSC's digital twin expertise with Korean SMEs' industrial additive manufacturing, inspection, and autoclave capabilities, the project establishes a closed-loop validation framework that replaces trial-and-error parameter tuning with simulation-guided optimization. This joint R&D approach reduces scale-up risk, shortens development timelines, and enables reliable vacuum-integrity qualification of additively manufactured composite tooling for industrial deployment.
Key Words	1. Process Digital Twin 2. Machine Card / Material Card 3. Additive Manufacturing Tooling 4. Interlayer Fusion and Bead Quality 5. Vacuum Integrity

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (TRL 4)	End of the Project	Level (TRL 6)

R&D Objectives

R&D Objectives

The primary objective of this joint R&D project is to develop and validate a process digital twin–based optimization framework that enables vacuum-integrity–qualified additively manufactured composite tooling for autoclave applications.

Detailed Objectives

- Development of material cards capturing extrusion flow behavior, thermal properties, and interlayer fusion characteristics of composite feedstock materials.
- Development of machine cards representing printer-specific thermal boundaries, deposition behavior, and process variability.
- Integration of flow, thermal, and bonding simulations to predict bead geometry, overlap, and fusion quality.
- Establishment of simulation–experiment correlations linking process parameters to vacuum integrity performance.
- Demonstration of simulation-guided printing parameter optimization for vacuum-integrity–qualified tooling.

R&D Target Specifications

No.	KPI	Unit	Target Value	Measurement & Verification Method
1	Maximum autoclave operating temperature with maintained vacuum integrity	°C	≥ 180	Vacuum hold and decay testing during autoclave cycles
2	Maximum autoclave operating pressure with maintained vacuum integrity	bar	≥ 6	Vacuum integrity verification under pressurized autoclave conditions
3	Vacuum pressure retention during autoclave cycle	inHg	≥ 22	Vacuum drop and hold-time measurements
4	Reduction in printing parameter trial-and-error iterations	%	≥ 50	Comparison of number of experimental iterations before and after digital twin–guided optimization
5	Digital twin prediction accuracy for bead geometry	%	≥ 90	Comparison of simulated bead geometry with measured bead dimensions
6	Digital twin correlation accuracy for interlayer bonding strength	%	≥ 85	Comparison between predicted and experimentally measured interlayer bonding strength
7	Demonstration scale of vacuum-integrity–qualified tooling	-	Lab/Industrial	Successful laboratory-scale and industrial-scale tooling fabrication and validation

R&D Plan

Year 1: Material Card Development and Process Physics Characterization

- Material characterization to establish material cards, including extrusion flow behavior, thermal properties, and interlayer fusion characteristics.
- Experimental calibration of interlayer bonding models using controlled printing trials.
- Initial validation of bead geometry, bead overlap, and interlayer fusion trends under representative printing conditions.

	Year 2: Machine Card Development and Process Digital Twin Integration • Development of printer-specific machine cards, capturing thermal boundaries, deposition behavior, and process variability. • Calibration of thermal boundary conditions using in-process temperature and deposition measurements. • Flow, thermal, and bead overlap simulation under realistic printing conditions. • Integration of flow, thermal history, and interlayer fusion models into a coupled process digital twin. • Simulation-guided optimization of printing parameters targeting vacuum integrity performance. • Fabrication and validation of representative additively manufactured tooling at Purdue University. Year 3: Demonstration and Autoclave Validation • Full-scale fabrication of additively manufactured composite tooling in collaboration with Korean SME partners. • Verification of vacuum integrity through autoclave vacuum hold and decay testing under industrial operating conditions. • Correlation of digital twin predictions with experimental results from full-scale tooling. • Refinement of the process digital twin and development of process design guidelines for industrial deployment.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Composites Manufacturing and Simulation Center (CMSC) at Purdue University will serve as the primary technical lead and global knowledge authority for the development of the process digital twin framework. CMSC is internationally recognized as a leading center for composite additive manufacturing, simulation-driven manufacturing, and process–structure integration, with extensive experience in digital twin development for composite tooling and manufacturing systems. CMSC will architect and lead all core technical components of the process digital twin, including material card development, machine card modeling, and coupled flow, thermal, and interlayer fusion simulation. CMSC will establish the fundamental modeling methodologies, calibration strategies, and simulation–experiment correlation frameworks required to accurately predict bead geometry, interlayer bonding quality, and vacuum integrity of additively manufactured composite tooling. In addition, CMSC will drive simulation-guided printing parameter optimization, experimental calibration of digital twin models, and the development of quantitative vacuum-integrity evaluation methodologies. CMSC will also define standardized digital twin workflows, validation protocols, and best-practice guidelines, ensuring that the developed framework is robust, transferable, and directly applicable to industrial additive manufacturing and autoclave tooling environments.

Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the industrial implementation, validation, and scale-up of the process digital twin, with a primary focus on printing parameter optimization and vacuum integrity qualification under industrial conditions. The SME will provide industrial-scale additive manufacturing environments, execute simulation-guided printing trials, and generate high-fidelity experimental data required for calibration and validation of machine cards and material cards. The SME will also lead pilot-scale demonstrations to ensure industrial applicability of the developed framework. Additionally, the SME will lead the fabrication of additively manufactured composite tooling using industrial large-scale 3D printing systems. The SME will implement and evaluate printing parameters derived from the process digital twin, including nozzle temperature, printing speed, bead overlap, layer time, and thermal management strategies. The company will perform printing, machining, and post-processing of composite tooling, with in-process monitoring and dimensional inspection to assess bead quality, deposition consistency, and geometric accuracy. Fabrication and process data generated by the SME will support machine card calibration and digital twin validation. The SME will also be responsible for vacuum integrity evaluation and autoclave-based validation of the additively manufactured tooling. This includes autoclave-based composite part production using the developed tooling, verification of vacuum integrity through vacuum hold and decay testing during autoclave cycles, and non-destructive testing (NDT) to assess internal defects and structural integrity. Inspection and autoclave performance data generated by the SME will be used to validate digital twin predictions related to bead fusion quality, leakage behavior, and vacuum integrity under operational conditions.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to establish a process-level digital twin platform that directly removes the primary technical barrier limiting industrial adoption of additively manufactured composite tooling for autoclave applications: unreliable vacuum integrity. By replacing empirical, trial-and-error parameter tuning with physics-based, simulation-guided optimization, the project fundamentally transforms how composite tooling is designed, qualified, and manufactured. The resulting workflow will significantly reduce development time, cost, and manufacturing risk while enabling repeatable production of vacuum-integrity–qualified tooling under industrial autoclave conditions. Rather than relying on operator experience or vendor-specific heuristics, manufacturers will be able to predict and control bead fusion quality, leakage behavior, and vacuum performance prior to fabrication. These outcomes will position participating Korean SMEs as technology-leading solution providers in advanced composite manufacturing, capable of delivering autoclave-ready tooling enabled by digital twin–driven additive manufacturing. The international collaboration with Purdue University further reinforces technical credibility and alignment with global manufacturing standards, accelerating access to global OEMs, Tier-1 suppliers, and equipment manufacturers and supporting long-term participation in high-value composite manufacturing supply chains.

6. References

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